

DETAILED DESCRIPTION OF THE INVENTION — WIRELESS COMMUNICATION SYSTEMS AND DATA SYNCHRONIZATION ARCHITECTURES

The intraluminal physiologic monitoring device may incorporate wireless communication and synchronization architectures configured to transmit physiologic data, receive configuration parameters, and coordinate operation with external computing systems or additional sensing nodes.

In various embodiments, communication may occur using Bluetooth®, Bluetooth Low Energy (BLE), Wi-Fi, ultra-wideband (UWB), near-field communication (NFC), proprietary low-power radio protocols, acoustic communication, optical communication, or hybrid communication architectures. Communication may be continuous, periodic, event-triggered, or deferred based on power management considerations.

Certain embodiments may utilize a companion computing device such as a mobile phone, tablet, dedicated gateway, or wearable hub to relay data between the intraluminal monitoring device and remote computational environments. Such intermediary devices may provide time synchronization, storage buffering, encryption management, or computational assistance.

Data synchronization methods may include timestamping, clock drift compensation, packet sequencing, error correction, loss detection, and retransmission protocols. Synchronization may be important for correlating intraluminal physiologic signals with external device data within multi-node ecosystems.

Communication systems may support bidirectional control allowing remote configuration of sensing parameters, firmware updates, calibration commands, measurement scheduling, and activation of specific sensor modalities. Secure update mechanisms may include signed firmware, authenticated update channels, or hardware-backed security modules.

In certain embodiments, the device may store physiologic data locally during offline operation and transmit data in batches when connectivity becomes available. Local buffering may reduce power consumption and allow operation in environments where continuous wireless connectivity is undesirable or restricted.

Network architectures may support direct device-to-device communication between sensing nodes, hub-and-spoke topologies mediated by a companion device, or cloud-coordinated synchronization across multiple devices and users.

Alternative embodiments may incorporate wired communication interfaces including sealed conductive contacts, docking interfaces, or externalized connectors configured for clinical or laboratory deployments.

The communication and synchronization approaches described herein are illustrative rather than limiting. Any wireless or wired communication system capable of supporting physiologic data exchange, device configuration, and multi-node coordination is intended to fall within the scope of the present disclosure.